

2016 HKDSE Physics

Paper 1A Multiple Choice Questions

Original questions with official answers and English explanations. Correct answers are red; explanations are blue.

An asterisk marks extension content in the original paper.

Question 1

Some icy cold liquid is kept cold inside a vacuum flask. Which statements are correct?

- (1) The flask's cork stopper reduces heat gain from the surroundings.
- (2) The silver coating on the inner surface of the glass wall is a good reflector of infra-red.
- (3) The vacuum between the double glass walls reduces heat gain by radiation.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

Official answer: A

- (1) Cork is a poor thermal conductor and the stopper also limits convection through the opening.
- (2) Silver reflects infra-red radiation, reducing radiative heat gain.
- (3) A vacuum suppresses conduction and convection, not radiation. Therefore only (1) and (2) are correct.

Question 2

0.3 kg of water at temperature 50°C is mixed with 0.2 kg of ice at temperature 0°C in an insulated container of negligible heat capacity. What is the final temperature of the mixture?

Given: specific heat capacity of water = $4200 \text{ J kg}^{-1}^{\circ}\text{C}^{-1}$

Specific latent heat of fusion of ice = $3.34 \times 10^5 \text{ J kg}^{-1}$

A. -1.8°C

B. 0°C

C. 1.8°C

D. 3.0°C

Official answer: B

Heat released if the water cools to 0°C : $Q = mc\Delta T = 0.3 \times 4200 \times 50 = 63000 \text{ J}$.

Melting all the ice requires $ml = 0.2 \times 3.34 \times 10^5 = 66800 \text{ J}$, which is greater. Some ice remains, so the equilibrium mixture is at 0°C . A negative final temperature would be incompatible with liquid water and ice coexisting at equilibrium.

Question 3 *

When an ideal gas is heated from 25 °C to 50 °C, the average kinetic energy of the gas molecules will

- A. double.
- B. increase by 41%.
- C. increase by 8.4%.
- D. increase by 4.1%.

Official answer: C

Average molecular kinetic energy is proportional to absolute temperature.

Fractional increase = $(323 - 298)/298 = 0.0839$, or 8.4%. Celsius temperatures cannot be used in this ratio.

Question 4

The speedometer of a car shown below indicates the car's



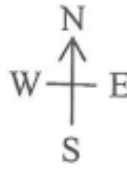
- A. instantaneous speed.
- B. instantaneous velocity.
- C. average speed of the whole journey.
- D. average velocity of the whole journey.

Official answer: A

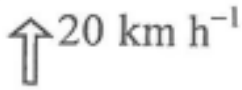
The instrument shows the magnitude of the car's velocity at that instant. It gives neither direction nor a journey average, so it measures instantaneous speed.

Question 5

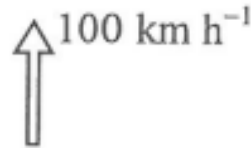
A car travelling at 80 km h^{-1} due east changes direction and travels at 60 km h^{-1} due north. Which diagram represents the change in velocity of the car?



A.



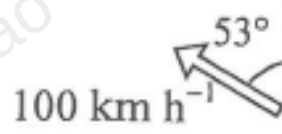
B.



C.



D.

**Official answer: D**

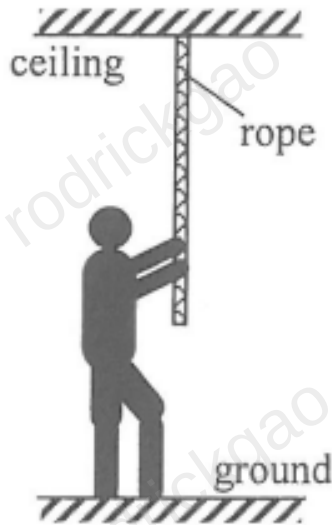
Take east and north as positive components. $\Delta v = v_{\text{final}} - v_{\text{initial}} = (-80, 60) \text{ km h}^{-1}$.

Magnitude = $\sqrt{(80^2 + 60^2)} = 100 \text{ km h}^{-1}$. The direction is 53.1° west of north because $\tan \theta = 80/60$. This is diagram D.

Question 6

A boy of weight W exerts a downward pulling force F on a rope of weight G hung vertically from the ceiling. He stands still on the ground as shown. Which of the following gives the magnitude of the force exerted by

- (1) the boy on the ground;
- (2) the rope on the ceiling?



- A. (1) W ; (2) $G - F$
- B. (1) W ; (2) $G + F$
- C. (1) $W - F$; (2) $G - F$
- D. (1) $W - F$; (2) $G + F$

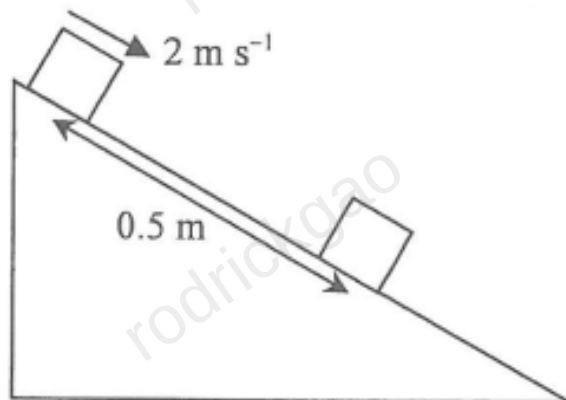
Official answer: D

The rope pulls the boy upwards with force F . Equilibrium of the boy gives $N + F = W$, hence the boy presses on the ground with $W - F$.

The ceiling supports both the rope's weight G and the boy's downward pull F , so the force on the ceiling is $G + F$.

Question 7

A block with initial speed 2 m s^{-1} slides down a rough inclined plane and stops after travelling a distance of 0.5 m . What is the deceleration of the block?



A. 1 m s^{-2}

B. 2 m s^{-2}

C. 4 m s^{-2}

D. Answer cannot be found as the angle of inclination of the plane is not given.

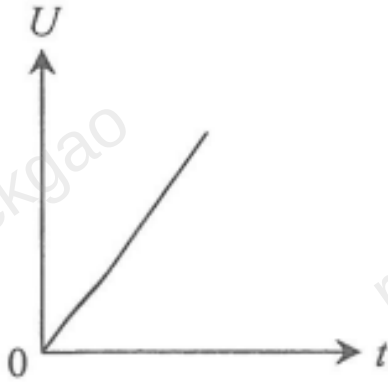
Official answer: C

Use $v^2 = u^2 + 2as$ along the slope: $0 = 2^2 + 2a(0.5)$, so $a = -4 \text{ m s}^{-2}$. The deceleration is its magnitude, 4 m s^{-2} ; the slope angle is unnecessary.

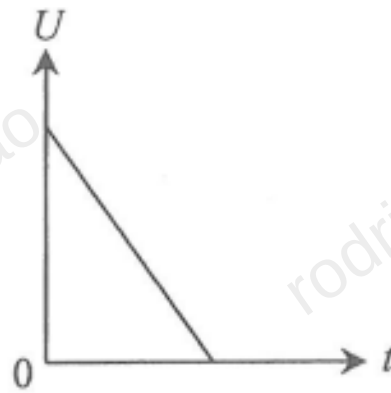
Question 8

An object at a certain height falls freely from rest under gravity. Which graph correctly shows the variation of its gravitational potential energy U with time t ? Neglect air resistance and take $U = 0$ at the ground.

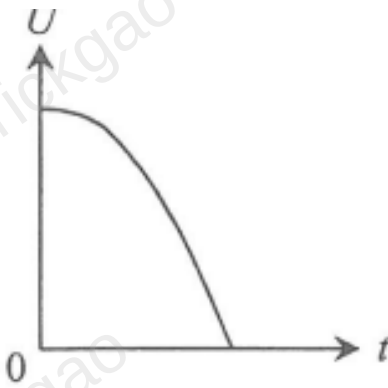
A.



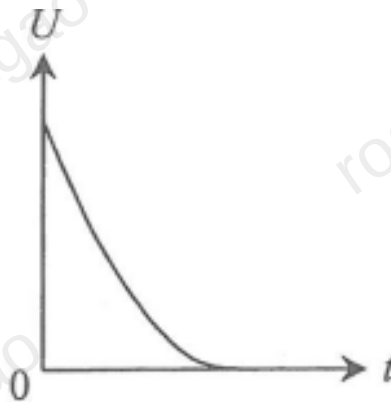
B.



C.



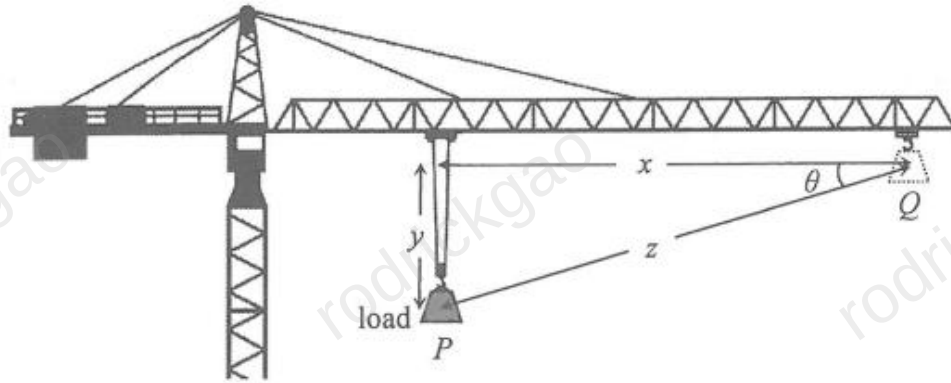
D.

**Official answer: C**

Before impact, $h = h_0 - \frac{1}{2}gt^2$ and $U = mgh = mgh_0 - \frac{1}{2}mg^2t^2$. Thus U starts above zero with zero initial gradient and decreases increasingly steeply, reaching zero at the ground: graph C.

Question 9

A crane moves a load of weight W steadily from point P to point Q as shown.
The work done on the load by the crane is



A. Wy .

B. $W(x + y)$.

C. Wz .

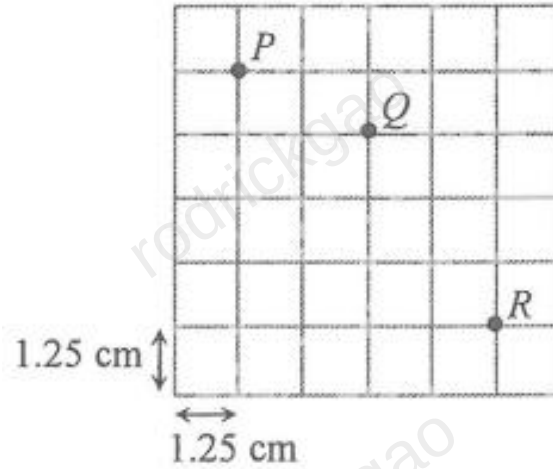
D. $Wz \cos \theta$.

Official answer: A

The load moves steadily, so its kinetic energy does not change. The crane's work equals the increase in gravitational potential energy, $W \times \text{vertical rise} = Wy$. The horizontal displacement does not add gravitational work.

Question 10 *

The stroboscopic picture shows a particle projected horizontally at position P into the air in a vertical plane. Subsequently the particle reaches positions Q and R such that the time interval between P and Q is equal to that between Q and R. Each square of the grid measures $1.25 \text{ cm} \times 1.25 \text{ cm}$. Find the particle's speed of projection at P. Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)



- A. 0.3 m s^{-1}
- B. 0.4 m s^{-1}
- C. 0.5 m s^{-1}
- D. 0.6 m s^{-1}

Official answer: C

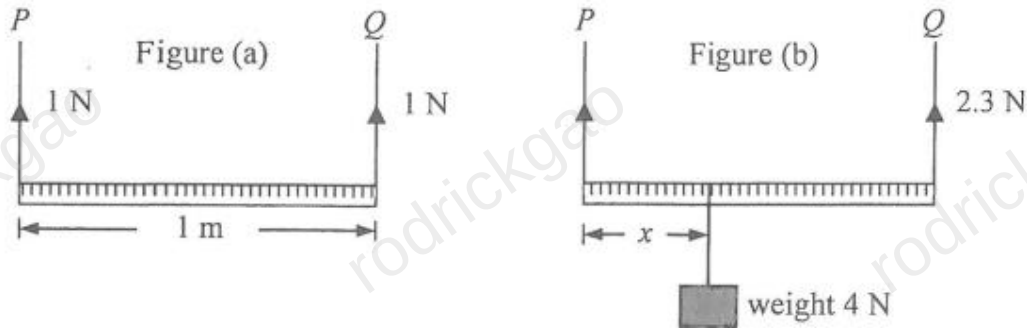
From P to Q, the horizontal displacement is 2 squares = 0.025 m and the vertical drop is 1 square = 0.0125 m .

The initial vertical velocity is zero, so $t = \sqrt{(2y/g)} = \sqrt{(2 \times 0.0125/9.81)} = 0.0505 \text{ s}$.

The horizontal velocity stays constant: $u = x/t = 0.025/0.0505 = 0.495 \text{ m s}^{-1} \approx 0.5 \text{ m s}^{-1}$.

Question 11

A uniform metre rule is supported by vertical wires P and Q and remains at rest horizontally as shown in Figure (a). The tension in each wire is 1 N. When a weight of 4 N is hung from the metre rule at a certain position as shown in Figure (b), the tension in Q becomes 2.3 N while the metre rule remains horizontal. Find the distance x shown.



- A. 32.5 cm
- B. 57.5 cm
- C. 67.5 cm
- D. Answer cannot be found as the tension in P is not known.

Official answer: A

The rule's weight is $1 + 1 = 2$ N, acting at its midpoint. Take moments about the left end in Figure (b):

$$2.3 \times 1 = 2 \times 0.5 + 4x.$$

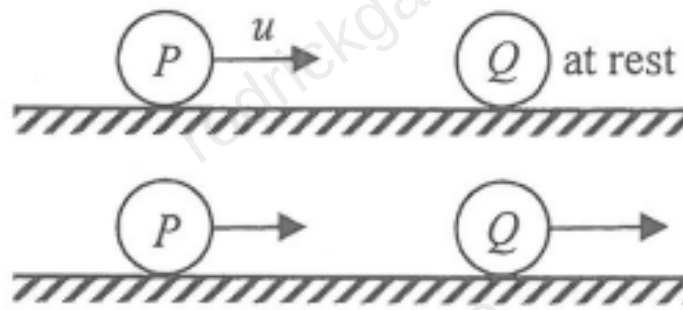
Hence $x = 0.325$ m = 32.5 cm. The unknown tension at P has zero moment about P.

Question 12

On a smooth horizontal surface, a marble P moving with speed u collides head-on with another marble Q, which is at rest. After collision, P and Q move with different speeds as shown.

Which of the following statements about this collision is/are correct?

- (1) During collision, the force acting on Q by P is equal and opposite to that acting on P by Q.
- (2) The total momentum of the two marbles is conserved only when the collision is perfectly elastic.
- (3) The kinetic energy lost by P must be equal to that gained by Q.



- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

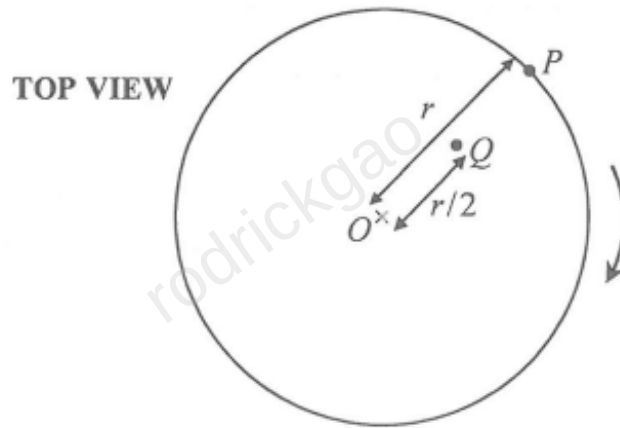
Official answer: A

- (1) The contact forces are a Newton's third-law pair.
- (2) Total momentum is conserved whenever the external impulse is negligible, including inelastic collisions.
- (3) Some kinetic energy may become internal energy, so P's loss need not equal Q's gain. Only (1) must be true.

Question 13 *

Particles P and Q are fixed at distances r and $r/2$ respectively from the centre O of a horizontal circular platform which is rotating uniformly as shown.

The ratio of the acceleration of P to that of Q is



A. 1 : 2.

B. 2 : 1.

C. 1 : 4.

D. 4 : 1.

Official answer: B

Both particles have the same angular speed ω because they are fixed to the same platform. Centripetal acceleration $a = \omega^2 r$, so $a_P/a_Q = r/(r/2) = 2$.

Question 14 *

A satellite orbits the Earth in a circular path of radius 7.2×10^6 m. What is the period of the satellite?

Given: mass of the Earth = 6.0×10^{24} kg

A. 1.4 hours

B. 1.7 hours

C. 1 day

D. Answer cannot be found as the mass of the satellite is not known.

Official answer: B

Gravity provides the centripetal force: $GMm/r^2 = m(4\pi^2 r/T^2)$.

$T = 2\pi\sqrt{r^3/GM} = 2\pi\sqrt{[(7.2 \times 10^6)^3/(6.67 \times 10^{-11} \times 6.0 \times 10^{24})]} \approx 6070 \text{ s} = 1.69 \text{ hours}$. The satellite mass cancels.

Question 15

The figure shows a snapshot of a transverse wave which travels along a string. Which statement is correct?



- A. The wave is travelling to the left if particle P is moving upwards at this instant.
- B. Particles P and R are moving in the same direction at this instant.
- C. Particle Q is at rest at this instant.
- D. Particle R vibrates with an amplitude larger than that of particle Q.

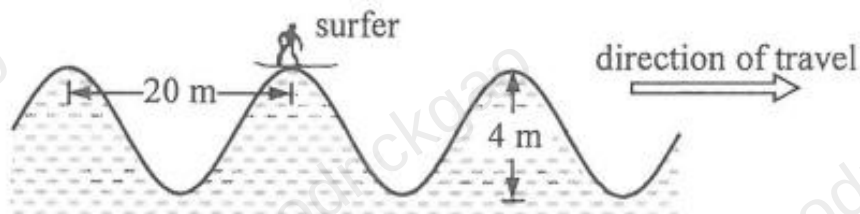
Official answer: B

P and R lie on portions with the same sign of slope, so their transverse velocities have the same sign for either propagation direction.

Q is crossing equilibrium, not at a turning point, and the wave amplitude is the same for all particles. If P moves upwards on the falling slope, the wave travels right, not left.

Question 16

The surfer in the figure reaches a crest at the moment shown. The crests of the water wave are 20 m apart and the surfer descends a vertical distance of 4 m from a crest to a trough in a time interval of 2 s. What is the speed of the wave?



- A. 1 m s^{-1}
- B. 2 m s^{-1}
- C. 5 m s^{-1}
- D. 10 m s^{-1}

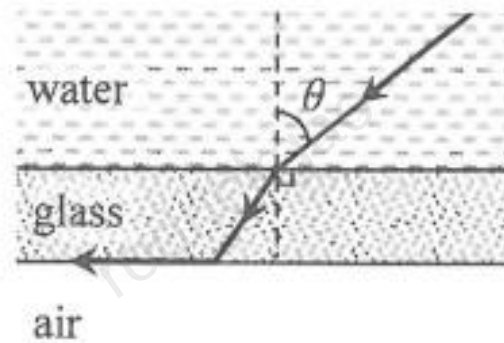
Official answer: C

Crest to trough takes half a period, so $T = 4 \text{ s}$. With wavelength $\lambda = 20 \text{ m}$, $v = \lambda/T = 20/4 = 5 \text{ m s}^{-1}$. The surfer's vertical distance is not the distance travelled by the wave.

Question 17

A parallel-sided glass sheet separates water from air. A ray of light in water is incident at an angle θ on the glass sheet and finally emerges into air along the glass-air interface as shown. Find θ .

Given: refractive index of water is 1.33.



- A. 41.2°
- B. 48.8°
- C. 53.1°
- D. It depends on the refractive index of glass.

Official answer: B

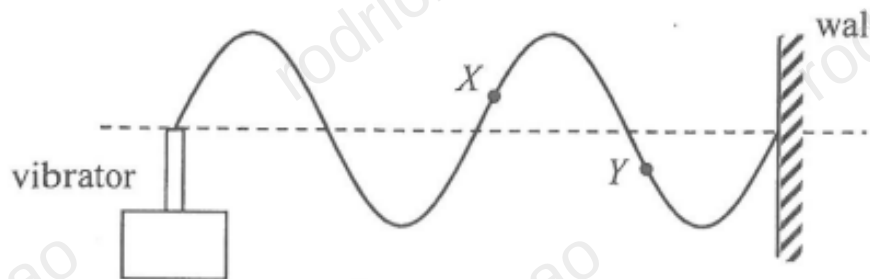
At the parallel interfaces, Snell's law gives $n_{\text{water}} \sin \theta = n_{\text{glass}} \sin r = n_{\text{air}} \sin 90^\circ = 1$.

Therefore $\theta = \sin^{-1}(1/1.33) = 48.8^\circ$. The intermediate glass index cancels.

Question 18

A string is tied to a vibrator while the other end is fixed to a wall. A stationary wave is formed as shown.

Which statement is correct when the frequency of the vibrator doubles?



- A. The wavelength will double.
- B. The wave speed will double.
- C. The amplitude will be halved.
- D. Particles X and Y will become vibrating in phase.

Official answer: D

For the same string tension, wave speed stays constant; doubling frequency halves wavelength and doubles the number of loops.

In the new mode, X and Y occupy loops separated by an even number of node boundaries, so they vibrate in phase. Frequency alone does not determine a halving of amplitude.

Question 19

Diffraction will occur when light

- (1) passes through a pinhole.
- (2) passes by a sharp edge.
- (3) passes through a slit.

- A. (1) only
- B. (2) only
- C. (3) only
- D. (1), (2) and (3)

Official answer: D

Diffraction is the spreading of a wave through an aperture or around an obstacle. It occurs at a pinhole, a sharp edge and a slit, even when the spreading is too small to notice easily.

Question 20

A beam of white light is separated into different colours after entering a glass prism because lights of different colours

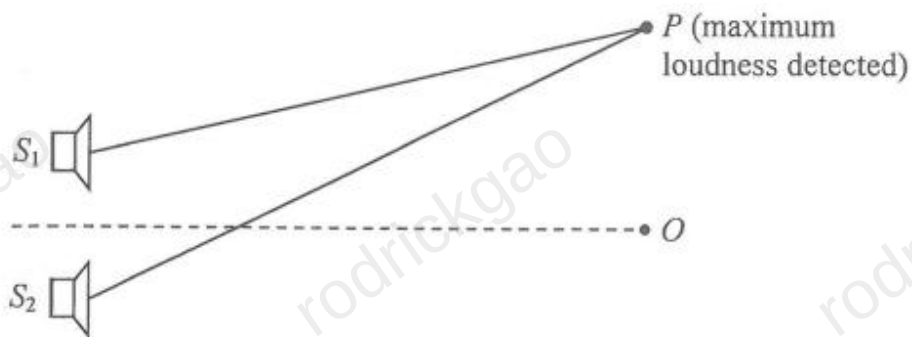
- A. are diffracted to different extents by the prism.
- B. undergo total internal reflection at different angles inside the prism.
- C. travel at different speeds in vacuum.
- D. travel at different speeds in glass.

Official answer: D

Different wavelengths have different refractive indices in glass. Since $v = c/n$, they travel at different speeds in glass and refract by different angles. In vacuum they all travel at c .

Question 21

Loudspeakers S_1 and S_2 connected to a signal generator emit sound waves which are in phase. Point O is equidistant from the loudspeakers while at point P maximum loudness is detected. The wavelength of the sound waves is λ . Which statement is INCORRECT?



- A. Both PS_1 and PS_2 must be integral multiples of wavelength λ .
- B. The definite value of the path difference $PS_2 - PS_1$ cannot be determined from the information given.
- C. At least one point of minimum loudness can be detected between O and P .
- D. Minimum loudness will be detected at P if the sound waves from S_1 and S_2 are in antiphase.

Official answer: A

Constructive interference requires the path difference $PS_2 - PS_1 = n\lambda$, not each path individually being an integral multiple of λ . Thus A is incorrect.

The non-central maximum at P has at least one intervening minimum from O ; changing the sources to antiphase exchanges maxima and minima.

Question 22

An object is moving at constant speed towards a convex lens of focal length 10 cm. At the moment when it is at 100 cm from the lens, which of the following descriptions of the image is correct?

- A. Direction: away from the lens; speed: faster than that of the object
- B. Direction: towards the lens; speed: faster than that of the object
- C. Direction: away from the lens; speed: slower than that of the object
- D. Direction: towards the lens; speed: slower than that of the object

Official answer: C

From $1/u + 1/v = 1/f$, $v = fu/(u - f)$. As u decreases while $u > f$, v increases: the image moves away from the lens.

At $u = 100$ cm, $v = 11.1$ cm. The image/object speed ratio is $(v/u)^2 = (1/9)^2 = 1/81$, so the image is slower.

Question 23

Which of the following are applications of ultrasound?

- (1) sterilizing drinking water
- (2) detecting cracks in railway tracks
- (3) breaking up kidney stones

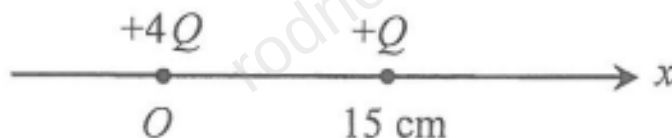
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: C

Ultrasound echoes locate cracks, and focused high-intensity ultrasound can break kidney stones. In the standard syllabus application, sterilizing drinking water uses ultraviolet radiation, not ultrasound. Thus (2) and (3) are the required choices.

Question 24 *

Point charges $+4Q$ and $+Q$ are fixed on the x -axis with $+4Q$ at the origin O and $+Q$ at $x = 15$ cm as shown. The respective electric fields due to the two charges are equal at



- A. $x = 10$ cm.
- B. $x = 12$ cm.
- C. $x = 20$ cm.
- D. $x = 30$ cm.

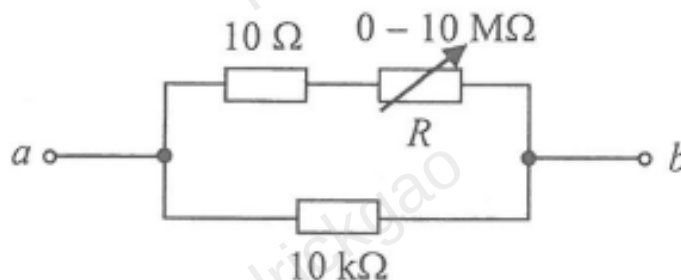
Official answer: D

Equal fields must have equal magnitude and direction. Between the positive charges the fields are opposite, so the zero-net-field point at 10 cm is not the answer.

To the right, $4/x^2 = 1/(x - 15)^2$. Hence $x = 2(x - 15)$, giving $x = 30$ cm; both fields point right.

Question 25

In the circuit, the variable resistor R can be adjusted over its full range from 0 to $10\text{ M}\Omega$. What is the approximate range of resistance between a and b ?



- A. 0 to $10\text{ k}\Omega$
- B. $10\text{ }\Omega$ to $10\text{ k}\Omega$
- C. $10\text{ }\Omega$ to $10\text{ M}\Omega$
- D. $10\text{ k}\Omega$ to $10\text{ M}\Omega$

Official answer: B

The resistance is $(10\text{ }\Omega + R)$ in parallel with $10\text{ k}\Omega$.

At $R = 0$, $R_{ab} = 10 \times 10000 / (10 + 10000) = 9.99\text{ }\Omega \approx 10\text{ }\Omega$.

At $R = 10\text{ M}\Omega$, the upper branch is much more resistive than $10\text{ k}\Omega$, so $R_{ab} \approx 10\text{ k}\Omega$.

Question 26

Two filament light bulbs X and Y are connected in parallel to a dry cell. X is brighter than Y . Which statements are correct?

- (1) In 1 s, the number of charges flowing through X is greater than that flowing through Y .
 - (2) In 1 s, the electrical energy dissipated by X is greater than that dissipated by Y .
 - (3) For every unit charge passing, the electrical energy dissipated by X is equal to that dissipated by Y .
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

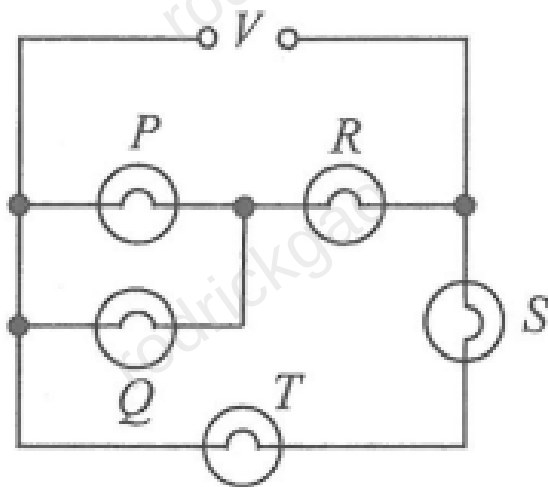
Official answer: D

Both bulbs have the same voltage because they are in parallel. The brighter bulb dissipates greater power, so $P_X > P_Y$ and $I_X > I_Y$ from $P = VI$.

Therefore more charge and more energy pass through X in each second. Energy per unit charge equals V and is the same for both. All three statements are correct.

Question 27

In the circuit, all the bulbs are identical. If the voltage V gradually increases, which bulb(s) will burn out first?



A. P and Q

B. R

C. S

D. T

Official answer: B

Let each bulb have resistance r . P and Q in parallel have resistance $r/2$, in series with R. Thus R receives $2V/3$ and each of P and Q receives $V/3$.

S and T are a separate series pair, each receiving $V/2$. R has the greatest voltage and power V_{bulb}^2/r , so it burns out first.

Question 28

A television set in stand-by mode consumes 1.5 W. If it is in this mode for 16 hours a day, estimate the carbon dioxide (CO_2) emission due to the electricity consumed in stand-by mode in a 30-day month.

Given: 1 kWh of electricity consumed corresponds to 0.8 kg CO_2 emission from the power station.

A. 0.576 kg

B. 0.720 kg

C. 576 kg

D. 720 kg

Official answer: A

Energy = $Pt = (1.5/1000) \times 16 \times 30 = 0.720$ kWh.

CO_2 emission = $0.720 \times 0.8 = 0.576$ kg. Convert W to kW before using hours.

Question 29 *

A student uses a search coil to study the strength of the magnetic field inside a long solenoid which is connected to an a.c. signal generator set at a certain frequency. Which of the following can improve the accuracy of this experiment?

- (1) Ensure the plane of the search coil is perpendicular to the field lines.
 - (2) Increase the signal generator's frequency and use the same current as before.
 - (3) Set the axis of the solenoid along an east-west direction to avoid the effects of the Earth's magnetic field.
- A. (1) only
 - B. (1) and (2) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

Official answer: Deleted

This item was deleted in the official marking scheme. No option is assigned as an official correct answer. It should not be treated as an ordinary scored question.

Question 30 *

A sinusoidal a.c. of a certain frequency delivers an r.m.s. voltage V_{rms} . If its frequency is doubled and its peak voltage is halved, what would be the r.m.s. voltage?

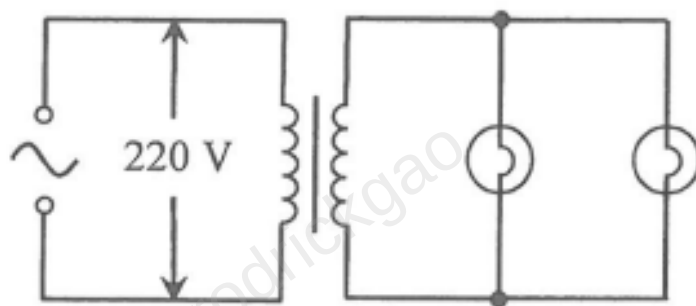
- A. $\frac{1}{2}V_{\text{rms}}$
- B. $V_{\text{rms}}/\sqrt{2}$
- C. $V_{\text{rms}}/(2\sqrt{2})$
- D. V_{rms}

Official answer: A

For a sinusoid, $V_{\text{rms}} = V_{\text{peak}}/\sqrt{2}$, independently of frequency. Halving the peak voltage therefore halves the r.m.s. voltage.

Question 31 *

In the circuit, each light bulb works at its rated value '22 W, 11 V'. The current in the primary coil is 0.25 A. Find the efficiency of the transformer.



- A. 20%
- B. 40%
- C. 64%
- D. 80%

Official answer: D

The two bulbs receive output power $2 \times 22 = 44$ W. Input power $= V_p I_p = 220 \times 0.25 = 55$ W.

Efficiency $= P_{\text{out}}/P_{\text{in}} \times 100\% = 44/55 \times 100\% = 80\%$.

Question 32

Which of the following statements about ionizing radiations is/are correct?

- (1) The ionizing power of α -particles is much stronger than that of β -particles.
- (2) γ -radiation can be completely shielded by a 10 cm thick concrete wall.
- (3) Ionizing radiations α , β and γ all undergo deflection in an electric field.

A. (1) only

- B. (1) and (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: A

(1) α -particles are strongly ionizing.

(2) Concrete attenuates γ -radiation; a finite thickness does not guarantee complete shielding.

(3) γ -rays are uncharged and are not deflected by an electric field. Only (1) is correct.

Question 33

Two radionuclides X and Y are of half-lives 3 hours and 4 hours respectively and initially there are N_X and N_Y undecayed nuclei respectively. After 24 hours, the number of undecayed nuclei of both nuclides becomes the same. Find $N_X : N_Y$.

A. 8 : 1

B. 4 : 3

C. 4 : 1

D. 2 : 1

Official answer: C

In 24 hours, X undergoes 8 half-lives and Y undergoes 6.

$N_X/2^8 = N_Y/2^6$, so $N_X/N_Y = 2^2 = 4$. Thus the initial ratio is 4 : 1.